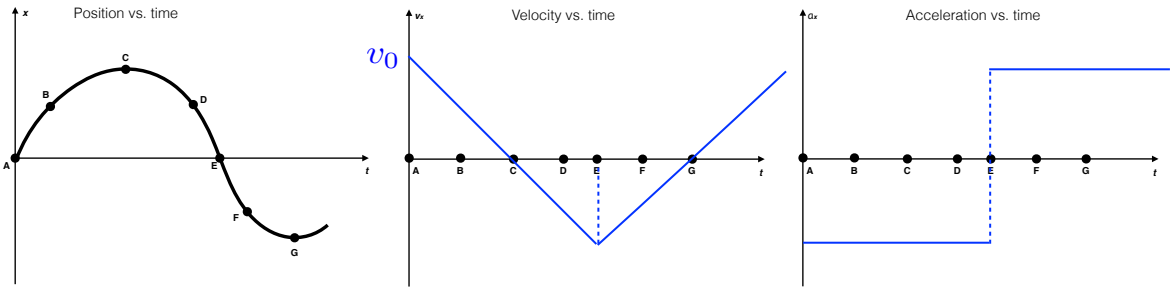
Conceptual Questions (30 points)

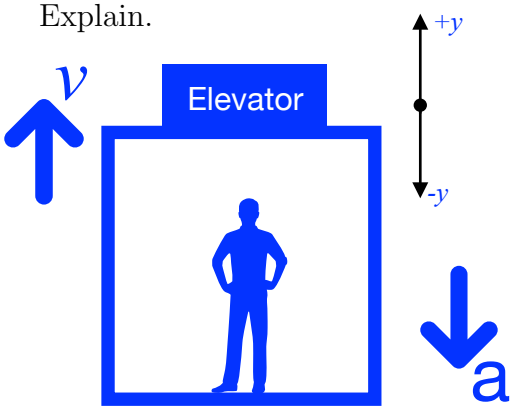
(a) (10 points) A ball thrown straight up has zero velocity at its highest point. Is the ball in equilibrium at this point? Why or why not? Explain.

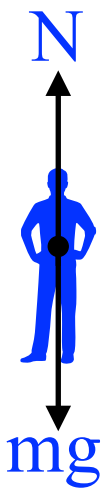
If the sum of forces acting upon on object is zero then the object is said to be in equilibrium. The weight of the ball (mg) acts on the ball even at the highest point. So, the ball is not in equilibrium.

(b) (10 points) An object is moving on the x-axis. Position vs. time graph at various time points is given below. The object (at $t = 0$) starts from the origin with positive x-velocity $\vec{v}_0$. Draw the x-velocity vs. time and x-acceleration vs. time graphs.



(c) (10 points) A woman weighs herself in an ascending (*i.e.* moving up) elevator with decreasing speed. How does her apparent weight compare to her actual weight? More, less or the same? Explain.





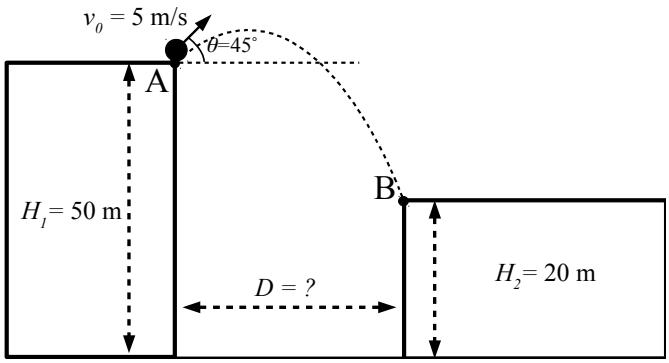
$$N - mg = m(-a)$$
$$N = m(g - a)$$

Less than the actual weight!

Problem (2) (30 points)

A projectile is launched from the edge of a building (point A) of height H_1=50 m with an initial velocity v_0=5 m/s that is θ=45° above the horizontal. If the projectile lands on the edge of another building of height H_2=20 m (point B), what should be the distance D between these two buildings?

(Take g=10m/s^2)



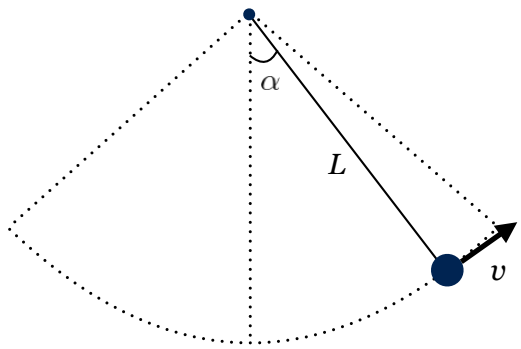
$$y(t) = y(0) + v_0 \sin(45°)t - \frac{1}{2}gt^2 = 50 + 5\frac{\sqrt{2}}{2}t - 5t^2$$

$$y(t') = 30 = 50 + 5\frac{\sqrt{2}}{2}t' - 5t'^2$$

$$t' = 2\sqrt{2}sec$$

$$D = v_0 \cos(45°)t' = 5(\frac{\sqrt{2}}{2})(2\sqrt{2}) = 10\ m$$

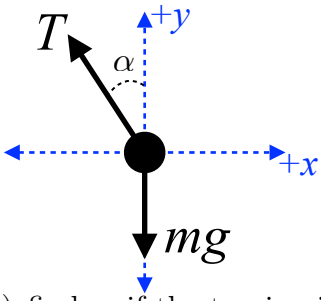
Problem (3) (30 points)



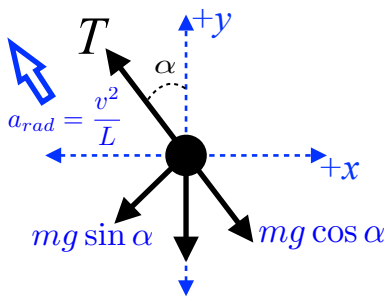
A bob of mass $m = 1.0$ kg is swinging at the end of a $L = 0.5$ m cord on a vertical plane. When the pendulum makes an angle of $\alpha = 36.9°$ with respect to the vertical the bob is moving with speed v in the direction shown. (Take g=10m/s^2)

<u>At this instant</u>,

(a) (10 pts.) draw the free-body diagram of the bob.



(b) (10 pts.) find v, if the tension in the cord is $T = 16$ N.



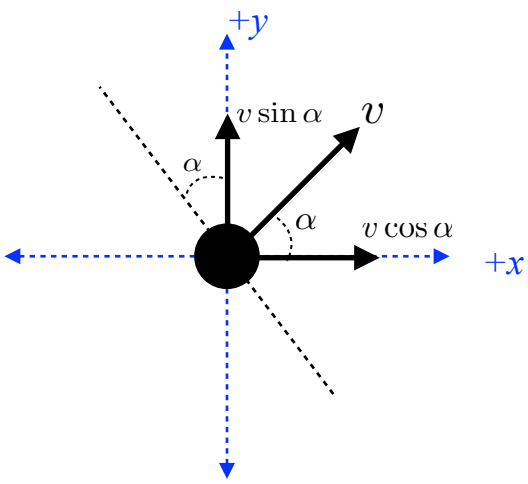
$$T - mg\cos\alpha = m\frac{v^2}{L}$$

$$16 - (1)(10)(\frac{4}{5}) = (1)\frac{v^2}{0.5}$$

$$v = 2\ m/s$$

v=

(c) (10 pts.) the cord breaks. Find the highest point the bob reaches above this level.



$$v_y(t) = v\sin\alpha - gt$$

$$v_y(t') = 0$$

$$(2)(\frac{3}{5}) - (10)(t') = 0$$

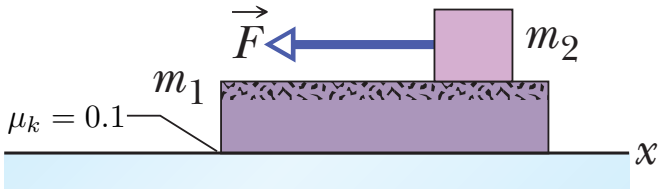
$$t' = 0.12\ sec$$

$$h_{max} = \frac{1}{2}gt'^2 = \frac{72}{1000}\ m = 7.2\ cm$$

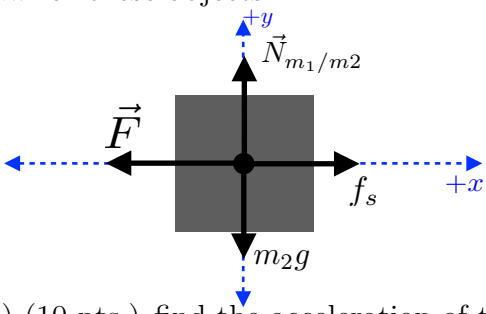
$h_{\max}$=

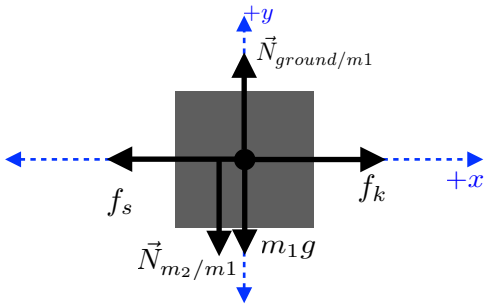
Problem (4) (30 points)

In the figure, a slab of mass m_1=40 kg rests on a floor where the coefficient of kinetic friction μ_k=0.1 between the slab and the floor, and a block of mass m_2=10 kg rests on top of the slab. Between block and slab, the coefficient of static friction is μ_s=0.95, and the coefficient of kinetic friction is μ_k=0.60. A horizontal force $\vec{F}$ of magnitude 100 N begins to pull directly on the block, as shown. If the object with mass m_2 is moving with the object with mass m_1 without sliding
(Take g=10 m/s^2)

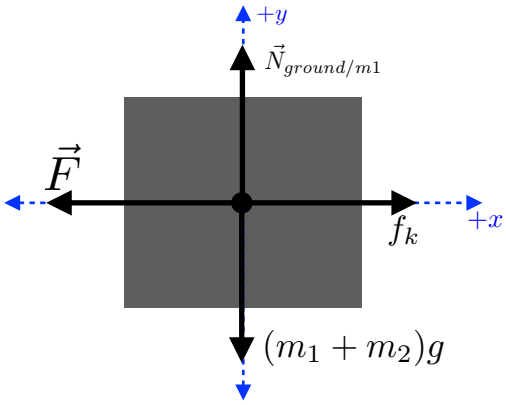


(a) (10 pts.) draw the free body diagrams for the slab and the block, and write Newton's Second Law for these objects.





(b) (10 pts.) find the acceleration of the system.



$$N_{ground/m1} = (m_1 + m_2)g = 500\ N$$
$$\mu_k N_{ground/m1} - F = ma_{system}$$
$$(0.1)(500) - 100 = 50a_{system}$$
$$a_{system} = -1.0\ m/s^2$$

negative sign means that the acceleration is in the same direction with F.

f=

(c) (10 pts.) find the friction force on the 10-kg block.

$$f_s - 100 = m_2 a_{system}$$
$$f_s - 100 = (10)(-1)$$
$$f_s = 90\ N$$

a=